ÉCOLE NATIONALE SUPÉRIEURE D'ARTS ET MÉTIERS
UNIVERSITÉ HASSAN II DE CASABLANCA

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TP N° 10 Machine Learning & Text Mining
SVM

Activité 1 : SVM Linéaire

Considérons les programmes suivants :

```
import pandas as pd; from numpy import *
import matplotlib.pyplot as plt
from sklearn.metrics import confusion_matrix
donnees= pd.read_csv("D:\\pointsSVM.csv",sep=';')
donnees=array(donnees); x=donnees[:,0:2]; y=donnees[:,2];
axes = plt.axes(); axes.grid()
plt.scatter(x[y ==+1][:, 0], x[y ==+1][:, 1], color='r')
plt.scatter(x[y ==-1][:, 0], x[y == -1][:, 1], color='b')
def f(X,Beta):    return Beta[0]+sum(X.T*Beta[1:])
def classer(X,Beta):    return int(sign(f(X,Beta)))
from sklearn.svm import LinearSVC, SVC
lsvm = LinearSVC(C=10000); X_train=donnees[:,0:2]; y_train=donnees[:,2]
lsvm.fit(X_train, y_train); score = lsvm.score(X_train, y_train)
print(score); Beta=list(lsvm.intercept_)+list(lsvm.coef_[0]); print(Beta)
plt.show()
```

```
axes = plt.axes(); axes.grid()
plt.scatter(x[y ==+1][:, 0], x[y ==+1][:, 1], color='r')
plt.scatter(x[y ==-1][:, 0], x[y == -1][:, 1], color='b')
Bleu=[]; Rouge=[]; milieu=[]
for abscisseX in range(1,13):
    y=-1.0*(Beta[0]+Beta[1]*abscisseX+1)/Beta[2];  Bleu+=[y]
    y=-1.0*(Beta[0]+Beta[1]*abscisseX-1)/Beta[2];  Rouge+=[y]
    y=-1.0*(Beta[0]+Beta[1]*abscisseX)/Beta[2];  milieu+=[y]
plt.plot(arange(1,13),Bleu,color='b')
plt.plot(arange(1,13),Rouge,color='r')
plt.plot(arange(1,13),milieu,color='g')
plt.show()
print(Beta)
print('[5,1]==>',f(array([5,1]),Beta));print('[2,1]',f(array([2,1]),Beta))
print('[6,9]==>',f(array([6,9]),Beta));print('[12,2]',f(array([12,1]),Beta))
```

Activité 2 : SVM non Linéaire

Considérons le programme suivant :

```
import pandas as pd; from numpy import *
import matplotlib.pyplot as plt
donnees= pd.read_csv("D:\\pointsSVM2.csv",sep=';')
donnees=array(donnees); x=donnees[:,0:2]; y=donnees[:,2];
axes = plt.axes(); axes.grid()
plt.scatter(x[y ==+1][:, 0], x[y ==+1][:, 1], color='r')
plt.scatter(x[y ==-1][:, 0], x[y == -1][:, 1], color='b')
def f(X,Beta):    return Beta[0]+sum(X.T*Beta[1:])
def classer(X,Beta):    return int(sign(f(X,Beta)))
from sklearn.svm import LinearSVC, SVC,NuSVC
X_tr=donnees[:,0:2]; y_tr=donnees[:,2]
lsvm= SVC(gamma='auto');lsvm.fit(X_tr, y_tr);
score = lsvm.score(X_tr, y_tr); print(score);
lsvm= NuSVC(gamma='auto');lsvm.fit(X_tr, y_tr);
score = lsvm.score(X_tr, y_tr); print(score);
lsvm= LinearSVC(C=1000);lsvm.fit(X_tr, y_tr);
score = lsvm.score(X_tr, y_tr); print(score);
Beta=list(lsvm.intercept_)+list(lsvm.coef_[0]); print(Beta)
plt.show()
```